Probability Homework #9 (Coverage: 8.1, 8.2, 8.3)

1. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} k(x^2 + y^2) & \text{if } (x, y) = (1, 1), (1, 3), (2, 3) \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant k, (b) the marginal probability mass function of X and Y, and (c) $E(X)$ and $E(Y)$.

(a) $\sum_{x,y \in \Delta} p(x,y) = 2k + 10k + 13k = 1$

$k = \frac{1}{25}$ #

(b) $p_X(x) = \begin{cases} \frac{12}{25} & \text{if } x=1 \\ \frac{13}{25} & \text{if } x=2 \end{cases}$ $p_Y(y) = \begin{cases} \frac{2}{25} & \text{if } y=1 \\ \frac{23}{25} & \text{if } y=3 \end{cases}$

(c) $E(X) = \sum_{x \in \Delta} x \cdot p_X(x) = 1 \cdot \frac{12}{25} + 2 \cdot \frac{13}{25} = \frac{38}{25}$

$E(Y) = \sum_{y \in \Delta} y \cdot p_Y(y) = 1 \cdot \frac{2}{25} + 3 \cdot \frac{23}{25} = \frac{71}{25}$

2. Two fair dice are rolled. The sum of the outcome is denoted by X and the absolute value of the difference by Y. Calculate the joint probability mass function of X and Y and the marginal probability mass functions of X and Y.

D_1 \ D_2	1	2	3	4	5	6
1	2 (0)	3 (1)	4 (2)	5 (3)	6 (4)	7 (5)
2	3 (1)	4 (0)	5 (1)	6 (2)	7 (3)	8 (4)
3	4 (2)	5 (1)	6 (0)	7 (1)	8 (2)	9 (3)
4	5 (3)	6 (2)	7 (1)	8 (0)	9 (1)	10 (2)
5	6 (4)	7 (3)	8 (2)	9 (1)	10 (0)	11 (1)
6	7 (5)	8 (4)	9 (3)	10 (2)	11 (1)	12 (0)

x \ y	0	1	2	3	4	5	$p_X(x)$
2	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
3	0	$\frac{2}{36}$	0	0	0	0	$\frac{2}{36}$
4	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{3}{36}$
5	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	0	$\frac{4}{36}$
6	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{5}{36}$
7	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	$\frac{6}{36}$
8	$\frac{1}{36}$	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	$\frac{5}{36}$
9	0	$\frac{2}{36}$	0	$\frac{2}{36}$	0	0	$\frac{4}{36}$
10	$\frac{1}{36}$	0	$\frac{2}{36}$	0	0	0	$\frac{3}{36}$
11	0	$\frac{2}{36}$	0	0	0	0	$\frac{2}{36}$
12	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{36}$
$p_Y(y)$	$\frac{6}{36}$						

3. Let the joint probability density function of random variables X and Y be given by

$$f(x,y)=\begin{cases}2 & \text{if } 0\le y\le x\le 1\\ 0 & \text{elsewhere.}\end{cases}$$

(a) Calculate the marginal probability density functions of X and Y, respectively.

(b) Find $E(X)$ and $E(Y)$.

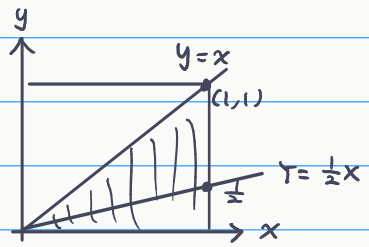
(c) Calculate $P(X<1/2)$, $P(X<2Y)$, and $P(X=Y)$.

(d) Are X and Y independent? Why or why not?

(e) $f_{X|Y}(x\mid y)=?$

(a) $f_X(x)=\int_0^x 2\,dy = 2x$

$f_Y(y)=\int_y^1 2\,dx = 2-2y$



(b) $E(X)=\int_0^1\int_0^x x\cdot 2\,dy\,dx=\int_0^1 2x^2\,dx=\frac{2}{3}$

$E(Y)=\int_0^1\int_y^1 y\cdot 2\,dx\,dy=\int_0^1 2y-2y^2\,dy=\frac{1}{3}$

(c) $P(X<\frac{1}{2})=\int_0^{\frac{1}{2}} f_X(x)\,dx = x^2\Big|_0^{\frac{1}{2}}=\frac{1}{4}$ #

$P(X<2Y)$：面積法 Total $=1\times 1\times\frac{1}{2}=\frac{1}{2}$ Area $=1\times\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$

$\frac{1}{4}\div\frac{1}{2}=\frac{1}{2}$ #

$P(X<2Y)$：$\int_0^1\int_{\frac{x}{2}}^x 2\,dy\,dx=\int_0^1 x\,dx=\frac{1}{2}x^2\Big|_0^1=\frac{1}{2}$

$P(X=Y)$：一條線 $=0$

(d) No, since $f(x,y)\neq f_X(x)\cdot f_Y(y)$

(e) $f(x|y)=\frac{f(x,y)}{f_Y(y)}=\frac{2}{2-2y}$ if $0\le y\le x\le 1$

4. Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{7}x^2y & \text{if } (x, y) = (1, 1), (1, 2), (2, 1) \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

$p_X(x) = \begin{cases} \frac{3}{7} & \text{if } x=1 \\ \frac{4}{7} & \text{if } x=2 \end{cases}$ $\quad p_Y(y) = \begin{cases} \frac{5}{7} & \text{if } y=1 \\ \frac{2}{7} & \text{if } y=2 \end{cases}$

$p(1,1) = \frac{1}{7} \neq p_X(1) \cdot p_Y(1) = \frac{3}{7} \cdot \frac{5}{7} = \frac{15}{49}$, so No

5. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 2e^{-(x+2y)} & \text{if } x \geq 0, \quad y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2Y)$.

$h(x,y) = X^2 \cdot Y$

$E(h(x,y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y) \cdot f(x,y)$

$= \int_0^{\infty}\int_0^{\infty} x^2y \cdot 2e^{-(x+2y)}\, dx\, dy$

$= \int_0^{\infty} 2y \int_0^{\infty} x^2 \cdot e^{-(x+2y)}\, dx\, dy$

$= \int_0^{\infty} 2y \cdot e^{-(x+2y)} \cdot (-x^2 - 2x - 2)\Big|_0^{\infty}\, dy$

$= \int_0^{\infty} 2y \cdot e^{-2y} \cdot 2\, dy$

$= 4\int_0^{\infty} ye^{-2y}\, dy$

$= 4 \cdot e^{-2y} \cdot (-\frac{1}{2}y - \frac{1}{4})\Big|_0^{\infty}$

$= 4 \cdot \frac{1}{4} = 1$

转	微
$e^{(2y-x)}$	x^2
$-e^{2y-x}$	$2x$
e^{2y-x}	2
$-e^{2y-x}$	0

转	微
e^{-2y}	y
$-\frac{1}{2}e^{-2y}$	1
$\frac{1}{4}e^{-2y}$	0

6. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x,y) = \begin{cases} \frac{1}{25}(x^2+y^2) & \text{if } x=1,2, \quad y=0,1,2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $p_{X|Y}(x|y)$, $P(X=2 \mid Y=1)$, and $E(X \mid Y=1)$.

$$p(x|y) = \frac{p(x,y)}{p_Y(y)} = \frac{x^2+y^2}{5+2y^2} \quad \# \quad x=1,2 \quad y=0,1,2$$

$$p_Y(y) = \sum_{x \in A} p(x,y) = \frac{1}{25}(1+y^2+4+y^2) = \frac{1}{25}(5+2y^2)$$

$$P(X=2|Y=1) = p(2,1) = \frac{5}{7} \quad \#$$

$$E(X|Y=1) = \sum_{x \in A} p(x|1) \cdot x$$
$$= 1 \cdot p(1,1) + 2 \cdot p(2,1)$$
$$= \frac{2}{7} + \frac{10}{7} = \frac{12}{7} \quad \#$$

7. A point (X, Y) is selected randomly from the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.

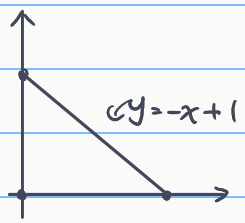
(a) Find the joint probability density function of X and Y.

(b) Calculate $f_{X|Y}(x|y)$.

(c) Evaluate $E(X \mid Y = y)$.

(a) Area of triangle $= \frac{1}{2}$

$$f(x,y) \begin{cases} \frac{1}{2} & \text{if } x \geq 0, y \geq 0, y < -x+1 \\ 0 & \text{otherwise} \end{cases} \quad \#$$



(b) $f_Y(y) = \int_0^{-y+1} \frac{1}{2} \, dx = \frac{1}{2}x \Big|_0^{-y+1} = \frac{1}{2}(1-y)$

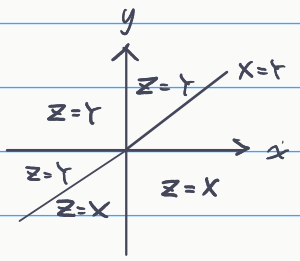
$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{1}{1-y} \quad \text{if } x \geq 0, y \geq 0, y < -x+1 \quad \#$$

(c) $E(X|Y=y) = \int_0^{-y+1} x \cdot \frac{1}{1-y} \, dx = \frac{1}{1-y} \int_0^{-y+1} \frac{1}{2}x^2 \Big|_0^{-y+1}$

$$= \frac{1}{1-y} \cdot \frac{1}{2} \cdot (1-y)^2 = \frac{1-y}{2} \quad \#$$

8. Let X and Y be two independent (continuous) uniform random variables chosen from the interval [-1, 1], respectively. Let Z= max (X,Y). Then (a) find the cumulative distribution function of Z, $F_z(t)$ = ? (b) compute the expected value of Z, E(Z).

$$F_z(t) = P(\max(x,y) \le t) = P(x \le t,\ y \le t)$$
$$= \int_{-1}^{t} f(x)\,dx \cdot \int_{-1}^{t} f(y)\,dy$$
$$= \int_{-1}^{t} \frac{1}{2}\,dx \cdot \int_{-1}^{t} \frac{1}{2}\,dy$$
$$= \left(\frac{1}{2}t + \frac{1}{2}\right) \cdot \left(\frac{1}{2}t + \frac{1}{2}\right)$$
$$= \frac{1}{4}t^2 + \frac{1}{2}t + \frac{1}{4} \quad \#$$



$$f_z(t) = \frac{1}{2}t + \frac{1}{2}$$

$$E(Z) = \int_{-1}^{1} z \cdot f(z)\,dz$$
$$= \int_{-1}^{1} z \cdot \left(\frac{1}{2}z + \frac{1}{2}\right) dz$$
$$= \int_{-1}^{1} \frac{1}{2}z^2 + \frac{1}{2}z\,dz$$
$$= \frac{1}{6}z^3 + \frac{1}{4}z^2 \Big|_{-1}^{1}$$
$$= \frac{1}{6} + \frac{1}{4} - \left(-\frac{1}{6} + \frac{1}{4}\right)$$
$$= \frac{1}{3}$$